Name: _____

WORKSHEET

# Slopes of curves

**1** Find the derivative of $f(x) = 4x^2$ and use it to determine the slope of the function $f$ at:

**a** $x = -3$

**b** $x = 0$

**c** $x = 1$

**d** $x = 1.5$

**e** $x = -6$

**f** $x = 4$

**2** Determine the slope of $y = -\frac{x^3}{3}$ at:

**a** $x = -2$

**b** $x = -1$

**c** $x = 0.5$

**d** $x = 3$

**e** $x = 10$

**f** $x = 100$

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**3** Find the gradient of $V = \frac{4}{3}\pi r^3$ at:

**a** $r = 0$

**b** $r = 1$

**c** $r = 2$

**d** $r = 4$

**4** Given $f(x) = \frac{1}{2}x^3$, find the value of the function and its slope to complete the table below.

| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $f(x)$ | | | | | | | |
| $f'(x)$ | | | | | | | |

**5** Given $y = 0.1x^4$, find the equation of the tangent at:

**a** $x = 0$

**b** $x = 1$

**c** $x = -3$

**d** $x = 5$

**6** Find the equation of the tangent to $p = \frac{1}{2}t^2$ at:

**a** $t = 0$

**b** $t = 1$

**c** $t = 3$

**d** $t = 8$

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**7** Find the equation of the tangent to the function $SA = 4\pi r^2$ at:

**a** $r = 1$

**b** $r = 4$

**c** $r = 0.5$

**d** $r = 2.5$

**8** Find the equation of the straight line which is perpendicular to the function $f(x) = 5x^2$ at:

**a** $x = 0$

**b** $x = -2$

**c** $x = 1$

**d** $x = 0.1$

**9** Find where the slope of $y = \frac{4}{3}x^3$ is 16.

**10** For what values of $z$ does the function $f(z) = z^5$ have a gradient of 3?

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## Answers

**1** **a** $-24$
**b** $0$
**c** $8$
**d** $12$
**e** $-48$
**f** $32$

**2** **a** $-4$
**b** $-1$
**c** $-0.25$
**d** $-9$
**e** $-100$
**f** $-10\,000$

**3** **a** $0$
**b** $4\pi$
**c** $16\pi$
**d** $64\pi$

**4**

| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $f(x)$ | $-13.5$ | $-4$ | $-0.5$ | $0$ | $0.5$ | $4$ | $13.5$ |
| $f'(x)$ | $13.5$ | $6$ | $1.5$ | $0$ | $1.5$ | $6$ | $13.5$ |

**5** **a** $y = 0$
**b** $y = 0.4x - 0.3$
**c** $y = -10.8x - 24.3$
**d** $y = 50x - 187.5$

**6** **a** $p = 0$
**b** $p = t - \frac{1}{2}$
**c** $p = 3t - \frac{9}{2}$
**d** $p = 8t - 32$

**7** **a** $\text{SA} = 8\pi r - 4\pi$
**b** $\text{SA} = 32\pi r - 64\pi$
**c** $\text{SA} = 4\pi r - \pi$
**d** $\text{SA} = 20\pi r - 25\pi$

**8** **a** $x = 0$
**b** $y = 0.05x + 20.1$
**c** $y = -0.1x + 5.1$
**d** $y = x + 0.15$

**9** $x = \pm 2$

**10** $\pm\sqrt[4]{\frac{3}{5}} \approx \pm 0.880$

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